Task 8:

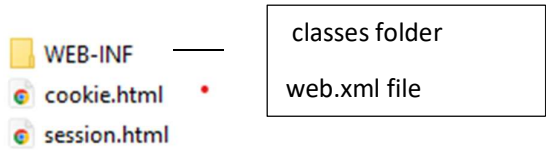
Maintaining the transactional history of any user is very important. Explore the various session tracking mechanism (Cookies, HTTP Session)

In tomcat folder

→ web apps

→ create ur application folder (e.g. task 8)

→ create



in the classes folder : all the java source files and javax.servlet api file should be there.

CookieLoginServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class CookieLoginServlet extends HttpServlet {

   public void doPost(HttpServletRequest req,
HttpServletResponse res)
      throws IOException {
```

```
        res.setContentType("text/html");

        String username = req.getParameter("username");
        String password = req.getParameter("password");

        // Create cookies
        Cookie userCookie = new Cookie("uname",
username);
        Cookie passCookie = new Cookie("pwd",
password);

        // Set cookie expiry (1 day)
        userCookie.setMaxAge(60*60*24);
      //  passCookie.setMaxAge(60*60*24);
         passCookie.setMaxAge(2*60);

        res.addCookie(userCookie);
        res.addCookie(passCookie);

        PrintWriter out = res.getWriter();
        out.println("<h2>Cookies Created
Successfully</h2>");
        out.println("<a href='retrieve'>retrieve cookie
details</a>");
    }
}
```

CookieRetrieveServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class CookieRetrieveServlet extends HttpServlet
{

   public void doGet(HttpServletRequest req,
HttpServletResponse res)
       throws IOException {

       res.setContentType("text/html");

       PrintWriter out = res.getWriter();

       Cookie[] cookies = req.getCookies();

       String username = "";
       String password = "";

       if (cookies != null) {
           for (Cookie c : cookies) {
               if (c.getName().equals("uname")) {
                   username = c.getValue();
               }
               if (c.getName().equals("pwd")) {
                   password = c.getValue();
               }
```

```
            }
        }

        out.println("<h2>Welcome " + username +
"</h2>");
        out.println("<h3>Password: " + password +
"</h3>");
    }
}
```

SessionLoginServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class SessionLoginServlet extends HttpServlet {

    public void doPost(HttpServletRequest req,
HttpServletResponse res)
        throws IOException {

        res.setContentType("text/html");

        String username = req.getParameter("username");
```

```
if (username != null && !username.isEmpty())
{

  // Create session
        HttpSession session = req.getSession();

//stores data in the session
session.setAttribute("uname", username);

        // Set session timeout = 3 months
        session.setMaxInactiveInterval(3 * 30 * 24 * 60 *
60);

        PrintWriter out = res.getWriter();
        out.println("<h1>Login Successful</h2>");
        out.println("<h2>session created
Successfully</h2>");

        out.println("<a href='retrieve1'> retrive session
details</a>");

    }

else {
            res.getWriter().println("Invalid Login!");
        }

}
}
```

SessionRetrieveServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class SessionRetrieveServlet extends HttpServlet
{

   public void doGet(HttpServletRequest req,
HttpServletResponse res)
       throws IOException {

       res.setContentType("text/html");

       PrintWriter out = res.getWriter();

//get current session
HttpSession session = req.getSession(false); // don't
create new
       if (session != null &&
session.getAttribute("uname") != null)
 {
//retrieve the data
String user = (String) session.getAttribute("uname");
if(session!=null)
{
 out.println("<h2>Welcome " + user + "</h2>");
               out.println("<a
href='LogoutServlet'>Logout</a>");
```

```
}
else {
            out.println("<h2>Session expired</h2>");
        }


}
else {
        out.println("<h2>No session found, login
first</h2>");
      }
   }
}
```

LogoutServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class LogoutServlet extends HttpServlet
 {
   public void doGet(HttpServletRequest req,
HttpServletResponse res)
        throws ServletException, IOException
{

res.setContentType("text/html");
```

```
        PrintWriter out = res.getWriter();

 HttpSession session = req.getSession(false);
        if (session != null) {
            session.invalidate(); // End the session

        }
out.println("<h2>Session closed. Redirecting...</h2>");
res.setHeader("Refresh", "3; URL=session.html"); //
Redirects after 3 seconds
    }
}
```

web.xml

```
<web-app>


<servlet>
        <servlet-name>cookie</servlet-name>
        <servlet-class>CookieLoginServlet</servlet-class>
    </servlet>


    <servlet-mapping>
        <servlet-name>cookie</servlet-name>
        <url-pattern>/login</url-pattern>
```

```
    </servlet-mapping>

    <!-- Retrieving Profile  -->
    <servlet>
        <servlet-name>Retrieve</servlet-name>
        <servlet-class>CookieRetrieveServlet</servlet-
class>
    </servlet>

    <servlet-mapping>
        <servlet-name>Retrieve</servlet-name>
        <url-pattern>/retrieve</url-pattern>
    </servlet-mapping>

<servlet>
        <servlet-name>session</servlet-name>
        <servlet-class>SessionLoginServlet</servlet-
class>
    </servlet>

    <servlet-mapping>
        <servlet-name>session</servlet-name>
```

```
        <url-pattern>/login1</url-pattern>
    </servlet-mapping>

    <!-- Retrieving Profile  -->
    <servlet>
        <servlet-name>SessionRetrieve</servlet-name>
        <servlet-class>SessionRetrieveServlet</servlet-
class>
    </servlet>

    <servlet-mapping>
        <servlet-name>SessionRetrieve</servlet-name>
        <url-pattern>/retrieve1</url-pattern>
    </servlet-mapping>
<servlet>
        <servlet-name>Sessionlogout</servlet-name>
        <servlet-class>LogoutServlet</servlet-class>
    </servlet>
<!--closing the session -->
<servlet-mapping>
        <servlet-name>Sessionlogout</servlet-name>
```

```
        <url-pattern>/LogoutServlet</url-pattern>
    </servlet-mapping>
</web-app>
```

cookie.html

```
<form action="login" method="post">
    Username: <input type="text"
name="username"><br><br>
    Password: <input type="password"
name="password"><br><br>
    <input type="submit" value="Login">
</form>
```

session.html

```
<form action="login1" method="post">
    Username: <input type="text"
name="username"><br><br>
    Password: <input type="password"
name="password"><br><br>
    <input type="submit" value="Login">
</form>
```